

EXPT NO: 7

DATA VISUALIZATION USING TABLEAU

AIM: To import, transform, analyze, and visualize data using Tableau Desktop / Public and create interactive dashboards and reports for decision-making.

SOFTWARE / TOOLS REQUIRED:

- Tableau Desktop / Tableau Public
- Dataset: OnlineRetail.csv

ALGORITHM:

1. Start
2. Open Tableau Desktop / Tableau Public.
3. Connect to the required dataset (OnlineRetail.csv).
4. Import and check the data fields and data types.
5. Clean and transform data by creating required calculated fields (e.g., Amount = Quantity * UnitPrice).
6. Create worksheets using suitable dimensions and measures.
7. Create visualizations such as bar charts, top-performer lists, and KPI summary tiles.
8. Combine the worksheets to create an interactive dashboard.
9. Add filters and slicers for dynamic analysis.
10. Analyze the dashboard and identify important business insights.
11. Stop.

PROCEDURE:

Step 1: Data Connection & Import

1. Launch Tableau Desktop / Tableau Public.
2. Under **Connect > To a File**, click **Text File** and select OnlineRetail.csv.
3. Review the data grid: ensure InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country columns are imported correctly.

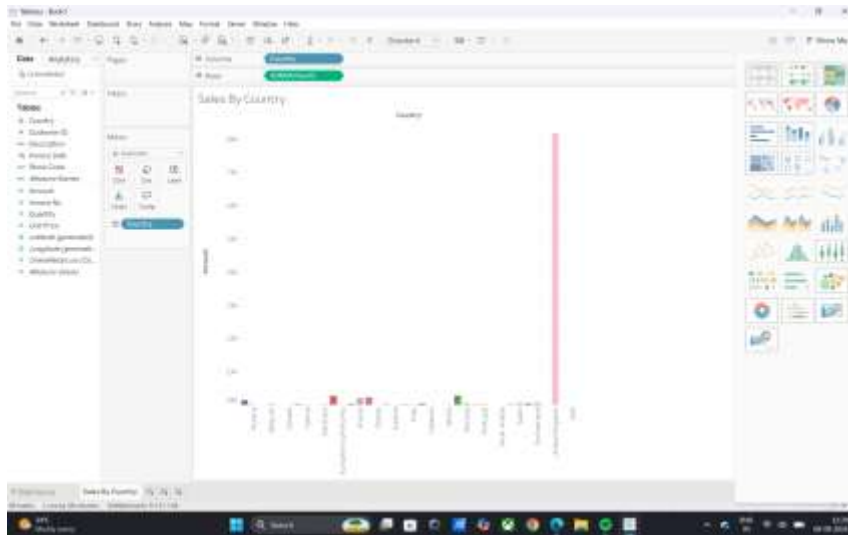
Step 2: Data Transformation (Calculated Field)

1. Click **Go to Worksheet (Sheet 1)**.
2. Right-click in the **Data Pane** on the left and select **Create Calculated Field**.

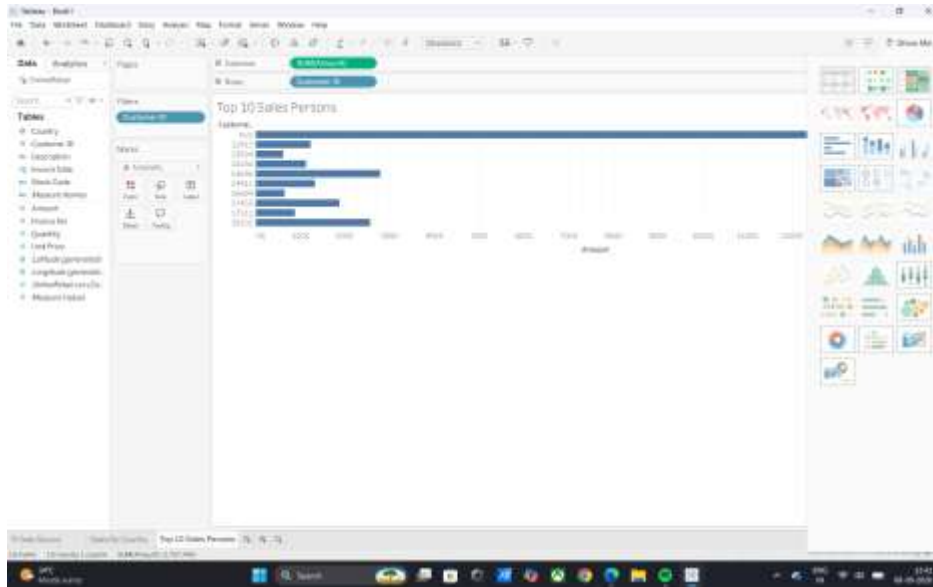
3. Set Field Name: Amount
4. Enter Formula: [Quantity] * [UnitPrice]
5. Click **OK**.

Step 3: Creating Visualizations (Worksheets)

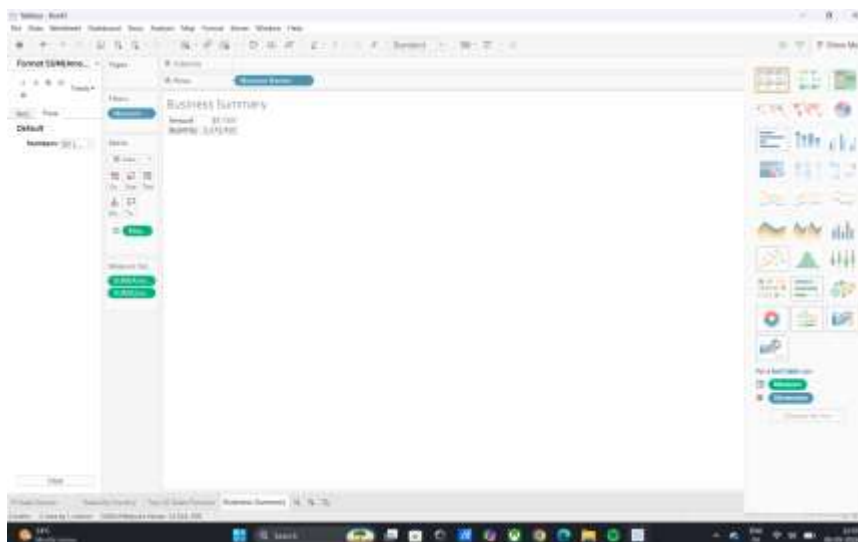
- **Worksheet 1: Sales by Country**
 1. Drag **Country** to **Columns**.
 2. Drag **Amount** to **Rows**.
 3. Drag **Country** to **Color** in the Marks Card.
 4. Rename sheet to Sales By Country.



- **Worksheet 2: Top 10 Customers**
 1. Create a new worksheet (**Sheet 2**).
 2. Drag **CustomerID** to **Rows**.
 3. Drag **Amount** to **Columns**.
 4. Drag **CustomerID** to **Filters** > select **Top** tab > set **By field: Top 10 by Amount (SUM)** > Click **OK**.
 5. Click **Sort Descending** on the top toolbar.
 6. Rename sheet to Top 10 Sales Persons.



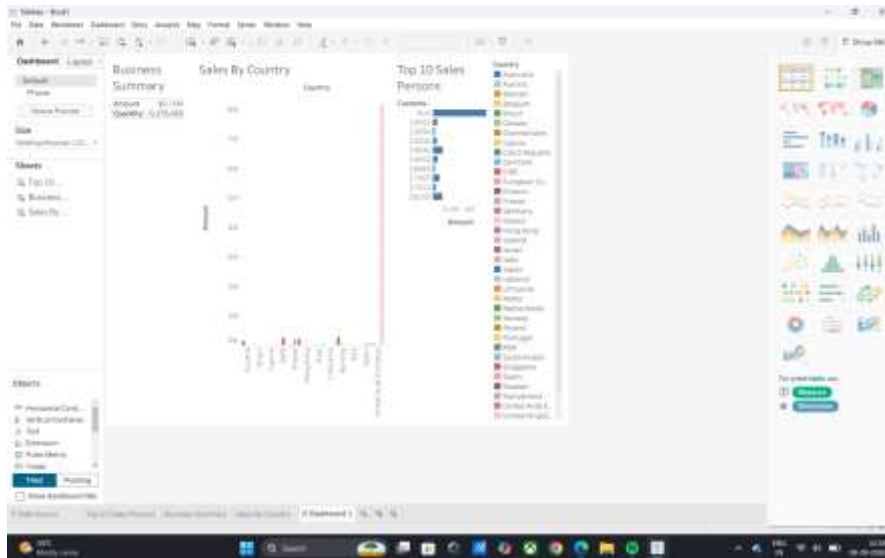
- **Worksheet 3: Business Summary KPI**
 1. Create a new worksheet (**Sheet 3**).
 2. Drag **Amount** and **Quantity** to **Text** on the Marks Card.
 3. Use **Show Me** to select **Text Table** format to display total metrics clearly.
 4. Rename sheet to Business Summary.



Step 4: Building the Interactive Dashboard

1. Click the **New Dashboard** icon at the bottom tab bar.
2. Set **Size** to **Automatic** in the left panel.
3. Drag Sales By Country, Top 10 Sales Persons, and Business Summary onto the canvas.

- Click on the Sales By Country sheet container and click the **Funnel icon ("Use as Filter")** to enable cross-filtering across all views.
- Check **Show dashboard title** and name it **Online Retail Sales & Analytics Dashboard**.



OUTPUT METRICS & OBSERVATIONS:

- Dataset Records Processed:** 541,909 rows
- Total Calculated Revenue (SUM(Amount)):** \$9,747,747.93
- Total Quantity Sold (SUM(Quantity)):** 5,176,450 units
- Top Revenue Region:** United Kingdom (\$8,187,806.36)
- Top Customer ID by Revenue:** Customer ID 14646.0 (\$279,489.02)

RESULT: The dataset was successfully imported, cleaned, analyzed, and visualized using Tableau. Different charts and an interactive dashboard were created to understand the data easily and support better decision-making.